

Event Memory with Addressable Sources

Evidence delivery, temporal questions
and transfer boundaries in multi-party dialogues

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Abstract

How can the changing facts of a long group conversation be made retrievable without losing their original source? We propose event memory with addressable sources (Elem): each event carries a self-contained retrieval description, a viewpoint owner, a subject, a time and immutable *source ID + verbatim quote* pairs. Memory is built before any question is read, events are ranked together with the original messages, and a retrieved event expands back into the primary messages.

The event index improves the delivery of annotated evidence relative to a strong dense RAW baseline. On 2,400 EverMemBench questions it raises the precision of gold sources by 3.21 pp (95% CI [2.84; 3.59]) and recall by 0.64 pp ([0.48; 0.80]), while returning on average fewer unique sources (9.47 versus 10.00) and more gold hits (2.245 versus 2.129). The direction holds in each of the five projects, and on SocialMemBench, with memory from an earlier extractor, the precision gain holds under resampling of conversation networks.

On the exploratory Temporal Duration slice ($n = 300$) events reach 19.3% accuracy against 14.0% for RAW and for HyDE at an approximately equal token budget and the same chronological order: the advantage persists after controls for volume, order and query expansion. The slice was selected post hoc, and after correction for the choice of category the signal does not reach confirmatory status.

The full system reaches 47.16% macro accuracy in an adapter evaluation of EverMemBench, above the published memory systems with the same answer model in a descriptive comparison; our own RAW reaches 46.34%. In paired comparisons, a mean accuracy gain from events and an added gain from linking into episodes are not established, and transfer across benchmarks is heterogeneous. The work shows that source delivery and answer quality should be evaluated separately: the event layer consistently improves the former, while its link to the latter remains an open question.

1 Introduction

Conversational memory faces a conflict between two goals. Retrieval is best served by a self-contained description: “Alexey said he will no longer come to the meeting.” Verification needs the original material: who exactly wrote this, when, and in what words. If the history is replaced by summaries, the answer becomes more accessible, but the boundary between evidence and interpretation disappears. If only messages are kept, the proof survives, but a late question has to guess anew all the elliptical, scattered and revised messages.

The problem is especially visible in a group. The author of a message, the owner of an opinion and the subject of a statement can be three different people. Several topics run at once, and positional proximity in the stream does not mean membership in the same line of conversation. Finally, states evolve: “I will be at the meeting” is replaced a week later by “I am leaving; I won’t come to the meeting.” A flat profile easily either keeps both versions as current or erases the first and deprives the system of any way to explain the change.

We propose to treat derived memory not as a new source of truth but as an *index over evidence*. In our implementation, Elem, an event contains (i) text optimized for semantic retrieval; (ii) a type, a viewpoint owner, a subject and an observed time; (iii) an immutable set of source IDs and literal quotes. Events compete with raw turns in a single vector top- k , and the selected derived documents expand back into messages. Episodes are an optional layer: they link revision, reaffirmation and augmentation, but do not replace events or their sources.

RETRIEVAL ABSTRACTION AND VERIFIABLE SOURCE

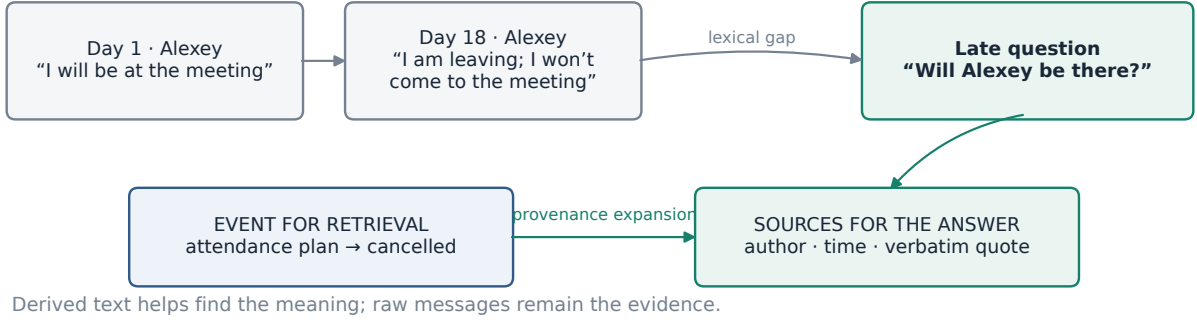


Figure 1: Two-view memory: a normalized event serves as the semantic bridge, while the answer is built from the returned primary messages. The example is illustrative and is not part of any benchmark.

The central question of this work: *does an event index help find evidence distributed across participants, messages and moments in time, while preserving access to the original sources?* Questions about duration and changes of state are a hard special case of this task. We answer in three steps: we measure the delivery of annotated sources on all questions, test the temporal advantage with controls for volume, order and query expansion, and determine what linking into episodes and transfer to other benchmarks add.

Contributions.

1. **Method.** An event index with addressable sources: the retrieval description is separated from the immutable support, each event carries a viewpoint owner, a subject, a time and *source ID + verbatim quote* pairs, and a retrieved event expands back into the original messages.
2. **Improved evidence delivery.** On all 2,400 EverMemBench questions the event index raises the precision and recall of delivered gold sources relative to a strong dense RAW baseline, while returning fewer unique sources and more gold hits; the direction holds in all five projects and is supported on SocialMemBench.
3. **Separate evaluation of delivery and answers.** The full system reaches 47.16% macro accuracy in an adapter evaluation of EverMemBench; the temporal advantage of events persists after controls for budget, order and HyDE; paired comparisons delimit the benefit: a mean accuracy gain from events and from linking into episodes is not established, and transfer across the three benchmarks is heterogeneous.

Memory, runs and statistics are sealed, and every number in the paper is recomputed offline (Appendix H).

2 Related work

Agent memory and conversational tasks. Generative Agents combine an experience stream, retrieval and reflection [Park et al., 2023]; MemGPT manages context tiers [Packer et al., 2023]. Mem0 consolidates facts [Chhikara et al., 2025], Zep/Graphiti uses a temporal graph [Rasmussen et al., 2025], A-MEM links notes [Xu et al., 2025], Hindsight separates facts, experience and beliefs [Latimer et al., 2025], and HippoRAG implements graph association [Gutiérrez et al., 2024]. Unlike a general-purpose memory manager, our evaluation focuses on multi-party conversation with stable participant identifiers. LoCoMo and LongMemEval study multi-session memory [Maharana et al., 2024, Wu et al., 2024]; EverMemBench, GroupMemBench and SocialMemBench add, respectively, long collaborative projects, reply structure and social attribution [Hu et al., 2026, Yang et al., 2026, Owolabi, 2026].

Closest retrieval representations. Dense X Retrieval indexes atomic self-contained propositions [Chen et al., 2023]. RAPTOR builds a hierarchy of summaries; in collapsed-tree mode original leaves and derived nodes are ranked together [Sarthi et al., 2024]. Self-contained statements and a joint pool of raw and derived documents are thus known. Elem carries them over to changing conversational states: an event explicitly separates the author, the opinion owner and the subject, carries an observed time and a per-fact *source ID*

+ *verbatim quote* pair, and the source address does not depend on the retrieval description or the episode version. Experimentally, we compare this construction with our own strong RAW baseline and separate the contributions of source delivery, context volume, order and episode linking; the work contains no direct comparison with proposition retrieval or with summaries carrying the same provenance (§10).

Query expansion and verifiability. HyDE generates a hypothetical document for the query embedding [Gao et al., 2023a]; event memory normalizes the corpus in advance. We compare the two on Temporal Duration with budget and order aligned. Like work on RAG and citation [Lewis et al., 2020, Gao et al., 2023b], we separate retrieval of support from answer quality: an existing pointer does not prove the semantic correctness of a paraphrase.

3 Problem statement

Let $M = (m_1, \dots, m_n)$ be the history of a single bounded conversation space. A message $m_i = (id_i, g_i, a_i, t_i, x_i)$ contains the space, the author, the time and the text. A builder B receives only M and creates memory $\mathcal{D}$ before QA opens. For a query q , retrieval selects documents $R_k(q, \mathcal{D})$, after which the generator answers from the expanded context C_q .

We distinguish three properties:

1. **pointer integrity**: the source ID exists, and the quote is literally contained in it;
2. **semantic support**: the source text actually entails the derived description;
3. **answer correctness**: the generator used the delivered context correctly.

A mechanical check guarantees only the first. The second requires an entailment or human audit; the third is measured by benchmark scoring.

4 Method

4.1 The immutable event

An event has the form

$$e = (z, o, s, k, r, \tau, P, L),$$

where z is the self-contained index text, o the viewpoint owner, s the subject, k the event type, r the temporal mode, τ the observed time, P the evidence and L the local context. Evidence consists of pairs $(id_i, quote_i)$; time is derived from the real timestamps of the sources.

Separating o from s lets “A called B unreliable” be represented as A’s opinion about B rather than as an unconditional fact about B; the correctness of these fields depends on extraction quality. The types cover state, opinion, decision, commitment, relationship, exception and the category other. A conversational reaction without long-term content may produce no event; a cap of at most two events per turn applies.

Example. An illustrative fragment of one working session, not from a benchmark:

10:02, Marina: “We move the release to Friday — Oleg’s payment module isn’t done.”
 10:05, Oleg: “I’ll finish it by tonight.”
 18:40, Oleg: “Done, ready to ship.”

A week later the question is asked: “Who held up the release, and did they finish their part?” The answer is contained in the 18:40 message, but it mentions neither the release, nor the payment module, nor the person concerned: its link to the question exists only through neighbouring messages. The schema admits two events:

- e_1 : z = “Marina tied moving the release to Friday to Oleg’s unfinished payment module”; o = Marina, s = Oleg, τ = 10:02; P : (10:02, “Oleg’s payment module isn’t done”).
- e_2 : z = “Oleg reported that he finished the payment module”; o = Oleg, s = payment module, τ = 18:40; P : (18:40, “Done, ready to ship”); the messages at 10:02 and 10:05 are in L as context resolving “done”.

The index text of e_2 makes the short confirmation retrievable by the words of the question, and the quote points to the original message. The same example shows the boundary of this guarantee. The quote confirms that the message exists and was written by Oleg at 18:40, but it does not confirm the interpretation “finished the payment module”: that rests on messages in L . Only the sources in P are returned to the generator together with the event; messages in L reach the context only if retrieval finds them independently. The explanatory message at 10:02 is therefore better included in P , and the literal quote check does not ensure this: it is semantic support (property 2 in §3), not pointer integrity. The fields play different roles: o distinguishes Marina’s statement *about* Oleg from Oleg’s own message, s specifies whom or what the fact is about, and τ separates the state “not done” from the later “done”. The example illustrates the role of the schema fields; measured effects are reported in §7.1–7.4.

4.2 Query-independent extraction and validation

The history is split along benchmark sessions: in EverMemBench by day and group; in GroupMemBench by channel/phase with blocks of up to 40 messages. Forced structured output permits an empty list. A deterministic validator then removes invented IDs, missing quotes, content-free values and invalid enums. Local context can help resolve a pronoun, but does not automatically become evidence.

4.3 Optional versioned episodes

An episode $E_j = ((e_1, \rho_1), \dots, (e_t, \rho_t))$ is an append-only line where each operation ρ is one of create, revise, reaffirm or augment. For a new event, candidates are searched only within the conversation network. The score is

$$S(e, E) = 0.6 \cos(\phi(z_e), \phi(A_E)) + 0.4 \cos(\phi(z_e), \phi(z_{last(E)})).$$

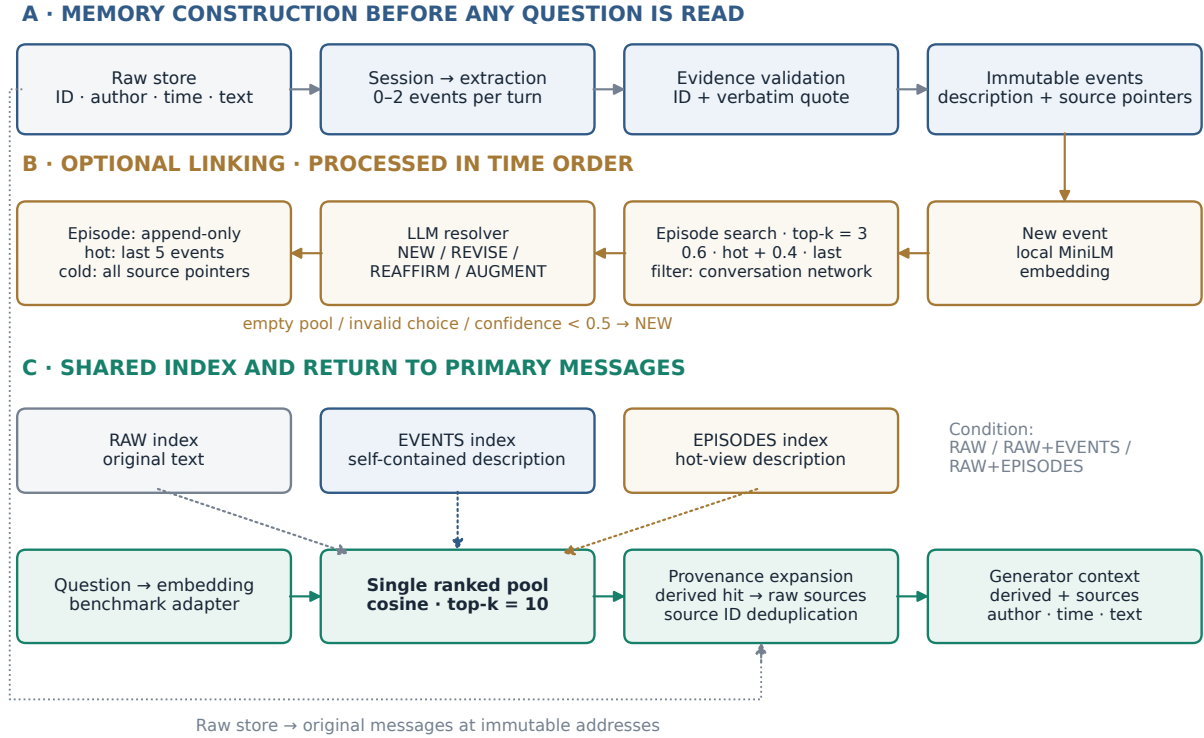
Here A_E is the concatenation of the last five events, $z_{last(E)}$ the last event, and ϕ a local MiniLM embedding for linking. The three best episodes go to an LLM resolver. With an empty pool, an invalid choice or confidence below 0.5, a new episode is created; this is a safeguard branch, not an additional search. Answer retrieval uses a separate embedding in the benchmark adapter. The index keeps a hot view of the last five events, while cold events and evidence are not deleted. We assume that a false split is cheaper than a false merge: split events can still reach retrieval independently, whereas an erroneous merge can bring extraneous sources into later requests; the frequencies of both errors were not measured.

4.4 Unified retrieval and provenance expansion

A raw turn and a derived document share a common interface

$$d = (id, kind, index_text, render_text, source_ids).$$

The question vector is compared against a single pool; the top-10 can contain both kinds. After ranking, each event/episode expands into unique raw turns. The generator sees the derived text as a navigational hint and the dated, speaker-grounded sources as the basis of the answer.



INVARIANT · the description may change; the primary history and the evidence address are preserved.

Figure 2: The full architecture. Memory is built query-independently and sealed before QA. In the main variant raw turns and episodes are ranked jointly; in the ablation, episodes are replaced by separate events. A derived hit always expands into raw sources.

Derived memory passes the same relevance criterion as raw history; static user profiles are not added.

5 Experimental protocol

5.1 Data and conditions

Benchmark	QA	Spaces	Main test
EverMemBench	2,400	5 projects	9 task types; temporal controls
SocialMemBench	1,031	43 networks	social attribution, norms, updates
GroupMemBench	745	4 domains	reply structure, implicit, multi-hop

Table 1: External datasets used. For GroupMemBench, 102 initially empty generation outputs were restored by a separate sealed amendment; the original protocol result is retained.

The main paired baseline is a strong dense RAW, not a weak keyword-only search. The EVENTS and EPISODES conditions add derived documents to the same top-10. For Temporal Duration the token budget and chronological order are also controlled; in the reordering controls the set of sources is fixed within each condition, but not between RAW and EVENTS. HyDE generates a hypothetical document only for the query embedding and returns the same raw documents. All variants use the same official answer/judge prompts inside the corresponding benchmark adapter.

EverMemBench uses GPT-4.1-mini for answering, Gemini-3-Flash-preview as the open-answer judge, text-embedding-3-large for retrieval and gpt-4o-mini-2024-07-18 for memory construction. The answer and judge models, their prompts and the exact multiple-choice scoring are taken from the official harness; providers are pinned (OpenAI and Google AI Studio respectively) with fallbacks disabled — this aligns the evaluation models, but not every other component of other systems. EverMemBench answers and judging

use temperature = 0. SocialMemBench follows the GPT-4o-mini official-compatible harness. GroupMemBench uses the GPT-5 of the official run; the provider’s model version was not pinned by date, so byte-level reproduction of these numbers is not guaranteed. Comparisons across benchmarks are not interpreted as comparisons of models.

5.2 Metrics and statistics

We report micro accuracy, macro accuracy over official question types, and paired differences between conditions. For a question q let U_q be the set of unique expanded sources and G_q the gold anchors released by the benchmark:

$$P_q = \frac{|U_q \cap G_q|}{|U_q|}, \quad R_q = \frac{|U_q \cap G_q|}{|G_q|}.$$

With an empty U_q precision is zero; with an empty G_q the scorer sets recall to one; in the verified Ever questions G_q is non-empty. Tables report *means of per-question ratios*, not the ratio of total hits to total sources. These metrics measure the delivery of annotated support, not entailment, context sufficiency or the correctness of user-facing citations.

95% CIs for Ever and Group come from a paired question bootstrap; for transfer to Social, 43 conversation networks are additionally resampled. Accuracy is compared with exact McNemar. Question-level intervals describe the studied set: its questions are not equivalent to independent projects. For evidence we therefore also report the effect per Ever project and a leave-one-project-out sensitivity (Appendix D).

Temporal Duration was selected after inspecting the full breakdown, so its analysis is marked exploratory; the correction for the choice of category is given in §7.2. Protocols, revisions and artifact sealing are in the appendix.

6 Results: system level and paired comparisons

Two questions are separated here. How strong the full system is in absolute terms is shown by a descriptive comparison with published rows (§6.1). What event memory itself adds is shown only by paired comparisons with our RAW on the same questions (§6.2) and by the controls in §7. The first answer characterizes the level the system achieves, the second the contribution of the event layer.

6.1 Comparison with published systems

EverMemBench. The full EPISODES system reaches **47.16% macro**, and RAW 46.34%. Our adapter result is above the values reported by Hu et al. [2026] with the same answer backbone: MemOS 42.55%, Zep 39.97%, Mem0 37.09%, Full Context 37.44% and MemoBase 34.27%. The gap to the best published memory system is 4.61 pp.

System	Macro	Multi-hop	Temporal	Update	Style
EPISODES (ours)	47.16	13.25	20.00	47.39	38.07
EVENTS (ours)	46.82	13.25	16.33	47.01	33.52
RAW (ours)	46.34	12.85	12.33	46.27	30.11
MemOS	42.55	18.88	15.67	45.15	28.98
Zep	39.97	8.03	13.00	43.66	26.70
Mem0	37.09	11.24	6.33	51.87	22.73
MemoBase	34.27	12.85	18.00	30.60	17.05

Table 2: Absolute EverMemBench level, accuracy in percent; answer model GPT-4.1-mini. Other systems: Table 4 of arXiv:2602.01313v3 [Hu et al., 2026]; our rows: full runs without token/order controls. Macro averages all nine types; the other columns show four diagnostic slices. There is no paired statistical test between the blocks.

On Temporal our EPISODES reaches 20.00%, against 6.33–18.00% for the published memory rows. The per-category profile is heterogeneous: MemOS is higher on Multi-hop, Mem0 on Update.

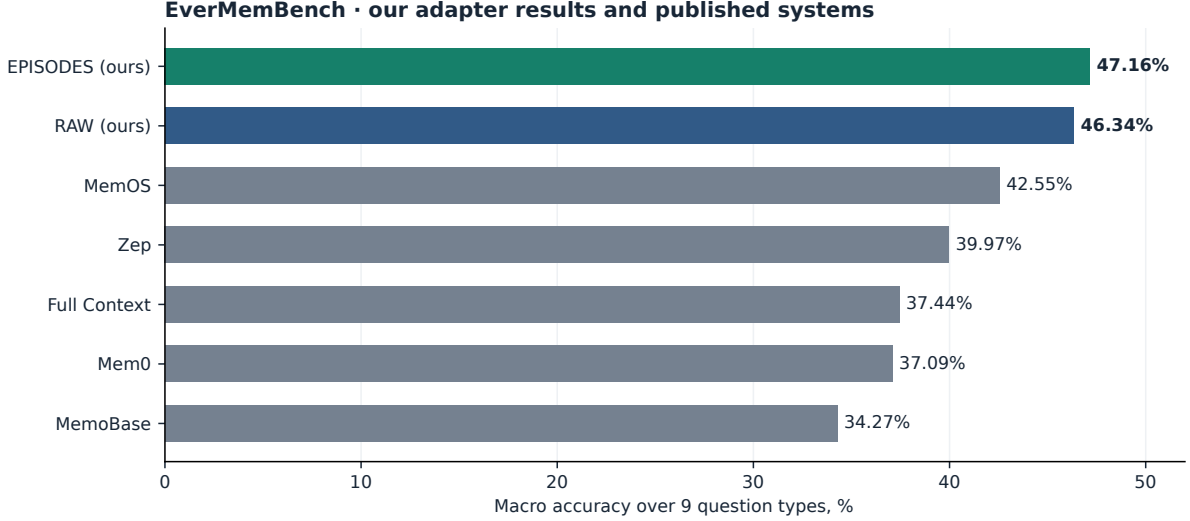


Figure 3: Macro accuracy over the nine EverMemBench categories with the GPT-4.1-mini answer backbone. Published results are from the benchmark paper [Hu et al., 2026]; our rows come from an adapter evaluation on top of the official harness. The comparison is descriptive: not an official leaderboard submission and not a paired test against other systems.

The comparison records the level achieved by the full adapter; other systems were not rerun on a single infrastructure. Our own strong RAW (46.34%) shows that most of this level comes from retrieval and the answer model, and the added contribution of memory is measured pairwise in §6.2.

SocialMemBench. On SocialMemBench the full system reaches MeanQ .368 (EPISODES) and .358 (RAW): practically at the level of the published full-context .369 and above the listed benchmark memory rows (best Subject-Mem .321).

System	MeanQ	Attribution	Decisions	ToM	State change
EPISODES (ours)	.368	.863	.691	.262	.259
RAW (ours)	.358	.825	.716	.251	.233
Subject-Mem	.321	.780	.610	.240	.220
SMG	.213	.440	.690	.110	.100
Graphiti	.178	.460	.610	.050	.070
LangMem	.162	.390	.560	.060	.090
Mem0	.143	.310	.570	.050	.070
Cognee	.120	.250	.590	.100	.010

Table 3: SocialMemBench: question-weighted scores (0–1), not binary accuracy. All rows use GPT-4o-mini. Other systems: Table 4 of [Owolabi, 2026]; ours: frozen official-v1 harness. Columns correspond to Q4/AP, Q2/GD, Q5/TM and Q8/TS. The comparison with external rows is descriptive.

The comparison exposes different demands on memory: the adapter achieves high attribution (.863 for EPISODES), while theory of mind and reconstructing state changes remain hard even with preserved sources. On group decisions EPISODES trail our RAW (.691 versus .716), so the high overall level reflects the whole adapter, not linking alone.

GroupMemBench. On GroupMemBench external results also depend on the task type (Appendix B). For example, published Hindsight reaches 54.94% on Temporal but 17.76% on Update; our EPISODES reaches 36.42% and 23.36% respectively. Published BM25 reaches 54.94% on Temporal and 25.23% on Update: comparing only against heavy memory systems would hide a strong simple alternative.

6.2 Paired comparisons within our conditions

Condition	Micro	Macro	Evid. recall	Evid. precision
RAW	49.46	46.34	8.08	21.29
EVENTS	49.83	46.82	8.71	24.50
EPISODES	50.21	47.16	10.75	22.45

Table 4: EverMemBench, all 2,400 QA, percent. Macro is the mean over the nine official question types; micro is over individual QA.

EverMemBench. At the micro level EPISODES improve 49.46%→50.21%: +0.75 pp, 95% CI $[-0.5; +2.0]$, $p = .256$. EVENTS reach 49.83%: +0.38 pp, CI $[-0.8; +1.6]$, $p = .589$. Both intervals include zero: a mean accuracy gain in the paired comparison is not established.

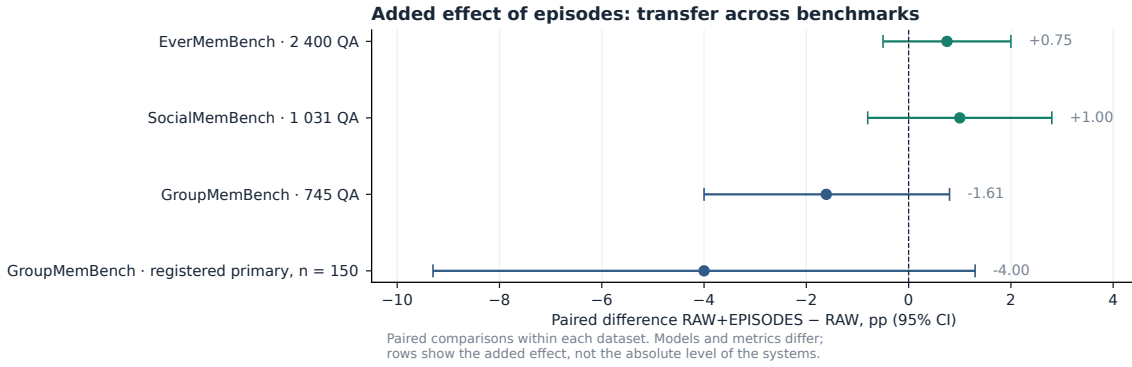


Figure 4: Paired differences EPISODES–RAW within each dataset with question- or network-bootstrap 95% CIs. Only paired differences are shown: the metric (accuracy versus MeanQ), protocols and answer models differ across the three datasets. Group is shown after the targeted repair of empty generations; its separately registered temporal/update endpoint ($n = 150$ of 745) is shown as its own row. No interval excludes zero.

Transfer across benchmarks. On SocialMemBench the paired difference EPISODES–RAW is +0.010 MeanQ, network-bootstrap 95% CI $[-.008; +.028]$: a mean MeanQ gain is not established.

At the level of delivery, Social supports the EverMemBench direction. The precision of delivered sources rises relative to RAW for flat memory (+0.90 pp, network-bootstrap 95% CI $[+0.66; +1.19]$), versioned (+0.86 pp, $[+0.61; +1.16]$) and episodes (+0.34 pp, $[+0.15; +0.54]$); all three intervals exclude zero. The result comes from a different conversation structure and a different answer model. The memory schemas are not identical: flat and versioned were built by an earlier extractor, and the base evidence recall on Social (67.5%) is substantially higher than on Ever, so absolute effect sizes are not comparable across datasets.

On GroupMemBench transfer is heterogeneous. On the pre-registered Group primary (Finance and Technology; knowledge update and temporal; $n = 150$), after restoring empty answers, RAW reaches 37.3% and EPISODES 33.3%: $\Delta = -4.0$ pp, CI $[-9.3; +1.3]$; over all 745 QA, 43.1% versus 41.5%, $\Delta = -1.6$ pp, CI $[-4.0; +0.8]$. The direction is negative and persists after the repair; the intervals include zero. Details of the 102 restored generations and of the original protocol result are in Appendix G.

7 The event representation: delivery and temporal answers

We first examine the direct change in retrieved sources, then its link to answer quality. This order evaluates the retrieval layer separately from the overall level of the answer model.

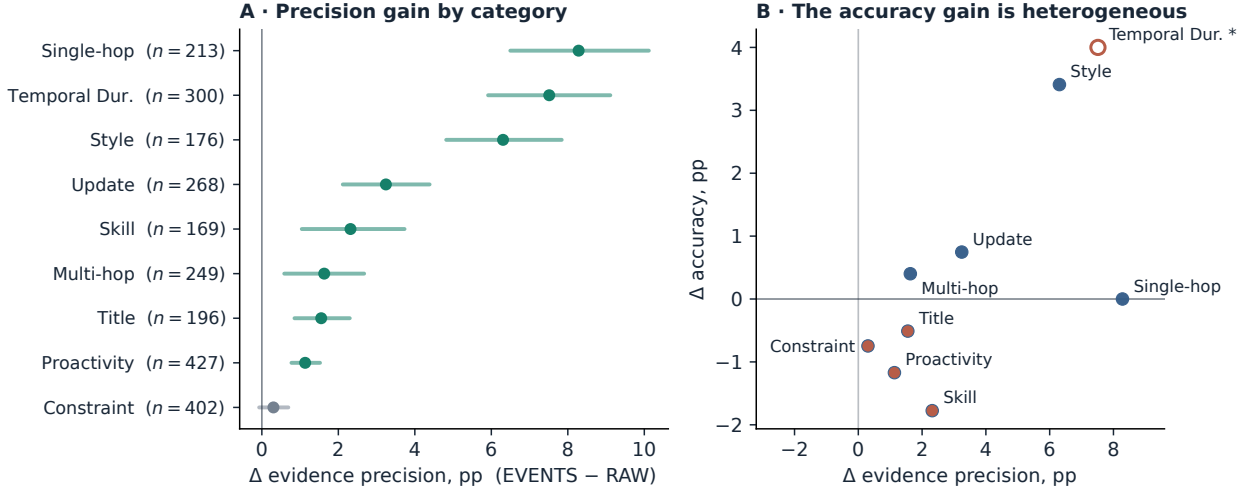


Figure 5: Evidence delivery and accuracy. **A:** paired difference in evidence precision EVENTS–RAW over the nine official categories, all 2,400 questions, question bootstrap 95% CI; the sign is positive everywhere, and the interval includes zero only for Constraint. **B:** the same axis against the shift in accuracy; the asterisk marks the post hoc discovery slice. The category with the largest delivery gain (Single-hop, +8.28 pp) sits at an accuracy ceiling of 89.7% and gains nothing. The points describe heterogeneity across categories, not a causal link between delivery and answers.

7.1 Evidence delivery: the main result

We compare, pairwise and per question, the share of delivered gold sources (recall) and their density in the context (precision) on all 2,400 questions, without slice selection.

Condition	Sources	Gold hits	Evid. precision
RAW	10.00	2.129	21.29
EVENTS	9.47	2.245	24.50
EPISODES	12.38	2.795	22.45

Table 5: EverMemBench, all 2,400 QA: how many unique sources entered the context, how many of them were gold anchors, and what share of the context was gold. On average EVENTS have fewer sources and more gold hits than RAW.

EVENTS–RAW, paired: precision +3.21 pp (95% CI [+2.84; +3.59], 855 questions better against 255 worse), recall +0.64 pp ([+0.48; +0.80]). The sign of the precision gain is positive in all nine official categories, and the CI excludes zero in eight of nine (Table 10). The effect does not rest on a single project: the gain is positive in each of the five EverMemBench projects, and leaving out any one project keeps it within 3.11–3.39 pp (Appendix D). The event index thus brings more annotated support into a fixed top-10.

The precision gain is not reducible to fewer returned sources. Precision could rise simply because fewer sources are returned. The data show otherwise: EVENTS return on average 9.47 sources against 10.00 for RAW, and *on no question* more, yet find more gold hits — 2.245 against 2.129 per question, a delta of +0.117 (CI [+0.083; +0.151]); on 449 questions gold hits are strictly higher without an increase in the number of sources. The event index improves the selection of support rather than merely shortening the list. In tokens the context is longer: the derived event text yields on average 1177 versus 1054 o200k_base tokens (recomputed offline from the sealed retrieval), so the role of volume in the temporal result is tested by a separate token control (§7.2). EPISODES behave differently: they raise recall together with volume (+2.68 pp recall at +2.4 sources), and their precision grows less (+1.16 pp).

Query-side expansion does not reproduce this. On Temporal Duration, at an approximately equal budget and the same order, matched HyDE obtains a similar recall — 20.7% against 19.7% for EVENTS (difference EVENTS–HyDE –1.01 pp, CI [–2.20; +0.17]) — but a markedly sparser context: the precision of EVENTS

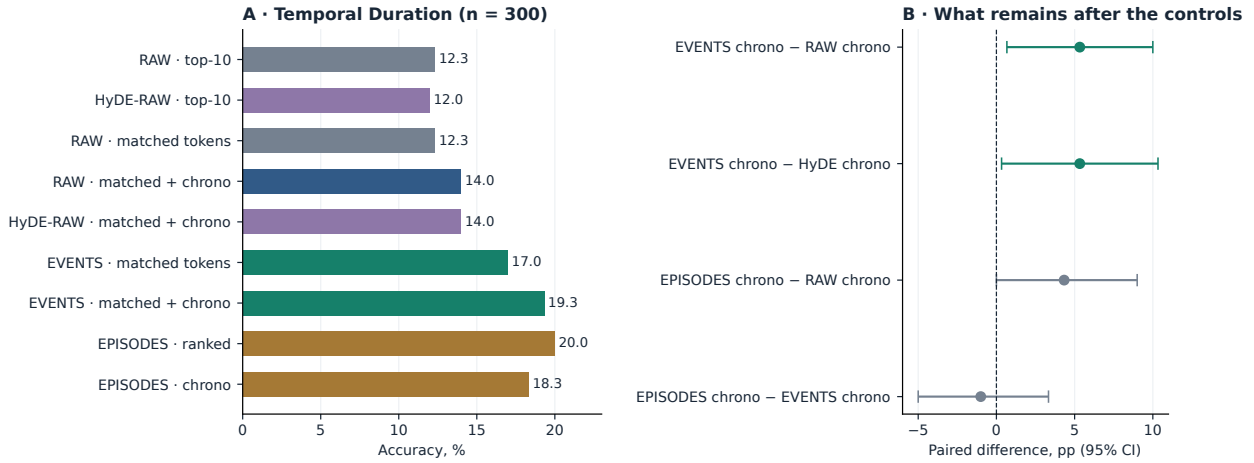


Figure 6: Controls for representation, volume and order on exploratory Temporal Duration. EVENTS chrono keep the observed advantage over RAW and HyDE at an approximately equal token budget and the same order; EPISODES do not exceed EVENTS. Intervals are question bootstrap 95%.

is higher by +6.68 pp (CI [+5.06; +8.27]). Against token-matched RAW, HyDE raises precision by +1.18 pp and recall by +1.61 pp. Query expansion and corpus normalization produce different delivery profiles: at a similar recall the event index delivers denser support. Dominance of EVENTS on both coordinates is not established: their recall is slightly lower, and a non-significant difference does not mean equivalence.

Why low recall is compatible with high accuracy. Ever gold annotation contains a median of 27 sources per question (interquartile range 21–37; maximum 96), while RAW returns ten. This sets a low completeness relative to the whole reference set. In RAW, 316 of 2,400 questions received a correct verdict without a single delivered gold anchor; in EVENTS, 312. Gold intersection and accuracy measure different properties of the system; the sufficiency of evidence for the answer was not evaluated separately.

7.2 Temporal Duration: the advantage after controls

In the original Temporal Duration breakdown RAW reaches 12.3% and EPISODES 20.0%. But episodes expand into several messages: on average 17.4 sources against ten for RAW. We therefore successively tested raw-context expansion, chronological order, separate events instead of linked episodes, and HyDE on the query side.

Token matching uses the token count of the expanded EPISODES context as the target budget and selects the closest prefix of ranked documents without truncating messages; this aligns budgets, it does not make them byte-identical. Matched HyDE targets the EVENTS context. Chrono controls change only the order of already selected blocks, keeping their sources. Mean chrono context sizes: RAW 2458, EVENTS 2456, HyDE 2451 tokens. The per-question absolute difference between EVENTS and RAW has a median of 35 tokens (1.48%), a 95th percentile of 97 (4.60%) and a maximum of 139 (11.46%). Full deviations and their computation rules are in Appendix E.

At an approximately equal token budget and the same order, EVENTS reach 19.3% against 14.0% for RAW: +5.33 pp, 95% CI [+0.67; +10.0], nominal $p = .0402$. Matched HyDE also obtains 14.0%; the difference EVENTS–HyDE is +5.33 pp, CI [+0.33; +10.33], nominal $p = .0479$ (37 wins against 21 losses). The direction is positive in all five projects. Outside Temporal Duration, on 2,100 questions, EVENTS give –0.14 pp (CI [–1.33; +1.10], $p = .877$): the advantage is concentrated in this slice rather than spread over all tasks.

The advantage comes with a characteristic delivery profile. Events combine the recall of expanded RAW (19.7% against 19.1%) with the precision of RAW top-10 (29.3% against 29.6%), whereas HyDE raises recall to 20.7% at 22.6% precision and 14.0% accuracy. This profile is consistent with the temporal advantage; the link between delivery and answers is tested in §7.3.

Condition	Accuracy	Evid. recall	Evid. precision
RAW top-10	12.3	12.7	29.6
HyDE-RAW top-10	12.0	14.1	32.4
RAW count-matched	13.3	16.6	25.1
RAW token-matched	12.3	19.1	21.5
RAW token-matched chrono	14.0	19.1	21.5
HyDE-RAW token-matched chrono	14.0	20.7	22.6
EVENTS token-matched	17.0	19.7	29.3
EVENTS token-matched chrono	19.3	19.7	29.3
EPISODES ranked	20.0	20.5	31.1
EPISODES chrono	18.3	20.5	31.1

Table 6: Temporal Duration, $n = 300$, percent. Precision and recall are means of per-question ratios after source expansion.

Status of the temporal signal. The slice was selected post hoc from nine categories. With a conservative Bonferroni correction the nominal $p = .0402$ and $.0479$ become about $.36$ and $.43$; with five projects the two-sided exact sign-flip test used cannot reach $p < .05$ (minimum $.0625$). Fixing the matched-HyDE endpoint in advance covers the analysis but not the choice of slice. The temporal result is thus a substantive exploratory signal, robust to controls for budget, order and HyDE, but not a confirmed effect.

7.3 Link between source delivery and answer quality

Before computing it, we recorded a rule for an offline check: outside Temporal Duration, split the questions by whether EVENTS or RAW dominate on recall and precision, and compare the direction of accuracy. In the stratum where EVENTS deliver strictly better support ($n = 650$), accuracy changes by $+0.77$ pp (CI $[-1.85; +3.54]$); where RAW dominates ($n = 209$), by -2.39 pp ($[-6.70; +2.39]$). The directions match the hypothesis, but the pre-registered confirmation criterion is not met (Appendix F).

This result separates what is established from what is open. The improvement in delivery is measured directly and consistently; its transfer to accuracy is not confirmed on the available samples. Evaluating the two endpoints separately shows which benefit the event layer already provides and which remains to be established. The check was run on a previously studied set, does not establish a causal mechanism and does not rule out an effect of delivery on answers: the interval in the stratum with better delivery also admits a noticeable positive effect.

7.4 What episodes add over events

In symmetric controls EPISODES chrono obtain 18.3% and EVENTS chrono 19.3% : a difference of -1.0 pp, CI $[-5.0; +3.33]$, $p = .755$. In the previously sealed comparison of ranked episodes (20.0%) and events chrono (19.3%), the difference was $+0.67$ pp, $p = .888$. An added mean accuracy gain from linking is not established.

Linking changes the delivery profile (§7.1) and keeps an inspectable chronology of revisions, reaffirmations and augmentations with access to earlier evidence. For retrieval in the studied scenario, separate events are the simpler basis: split events reach the context independently through ordinary ranking.

8 Efficiency and engineering properties

8.1 Cost and latency

The full EverMemBench experiment recorded 26,413 API calls and an upper list-price estimate of \$8.88. Of this, memory construction with gpt-4o-mini costs \$4.76, embeddings \$0.46, answers \$3.12 and the judge \$0.55. These figures refer to a research run with caches and are not a production TCO.

The total covers both conditions, repeats and answer scoring, not only EPISODES indexing, so the comparison with other systems below uses memory construction cost on GroupMemBench. The serialized episode snapshot occupies 12.77 MB; a full operational run together with the embedding cache is about 2.2 GB.

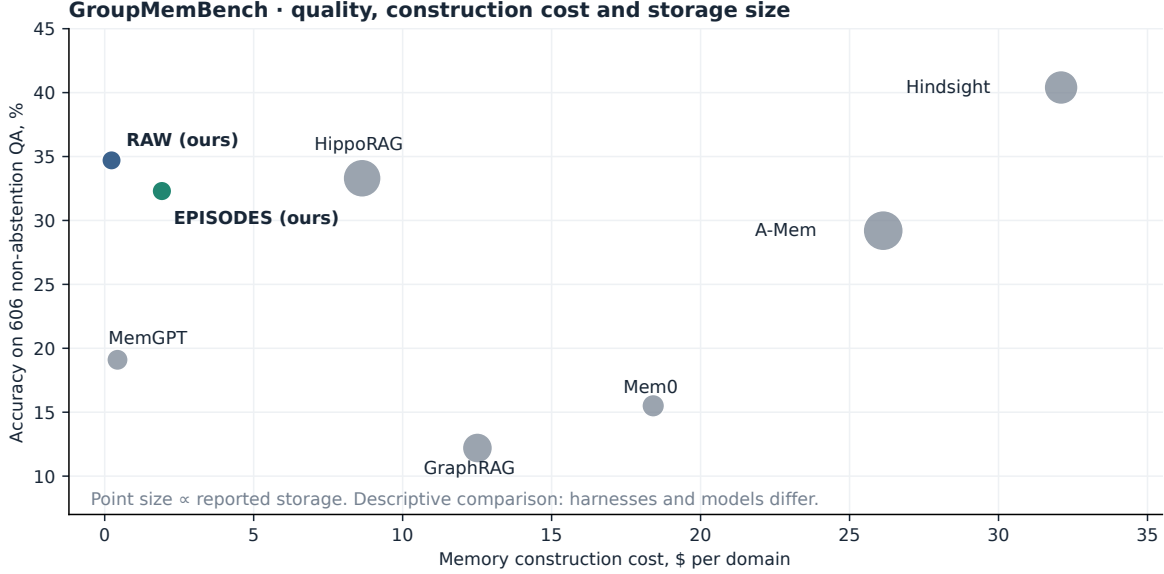


Figure 7: A descriptive GroupMemBench efficiency landscape on the shared sample of 606 QA, from the values of the benchmark paper and of our adapter. Point size encodes reported storage. Our points use the repaired amendment; cost definitions and backends differ, so the plot describes an engineering trade-off, not a controlled comparison of cost at equal quality.

System	Accuracy, %	Ingestion, \$/domain	Storage, GB
Hindsight	40.4	32.10	4.25
RAW (ours)	34.7	0.23	0.37
HippoRAG	33.3	8.64	5.76
EPISODES (ours)	32.3	1.92	0.41
A-Mem	29.2	26.13	6.60
MemGPT	19.1	0.43	0.69
Mem0	15.5	18.41	0.99
GraphRAG	12.2	12.51	3.00

Table 7: Explicit values for Figure 7. Accuracy is an aggregate over the five non-abstention task types of GroupMemBench (606 QA): for published systems it is recomputed from per-type percentages (Appendix B), for ours it is computed from the repaired results. It is a comparison of aggregates, not a single run of all systems. Ingestion cost and storage come from the corresponding benchmark reports. Our accuracies are recomputed after the targeted repair; the cost of the repair answers is not included in ingestion. Backends, harnesses and storage definitions differ, so the comparison is descriptive.

On this shared sample Hindsight reaches 40.4% at a reported ingestion of \$32.10/domain and 4.25 GB; HippoRAG 33.3%, \$8.64 and 5.76 GB; our RAW 34.7%, \$0.23 and 0.37 GB; EPISODES 32.3%, \$1.92 and 0.41 GB. The system is economical relative to heavy structural solutions, but Hindsight is more accurate. Because the harness components differ, this is context, not a controlled causal estimate of cost.

The adapter reaches accuracy at the level of HippoRAG without a large graph/database backend: by the published values, Hindsight’s construction costs about 16.7 times as much as our EPISODES, and its storage is 10.4 times larger; for HippoRAG these ratios are 4.5 and 14.0. These are ratios of reported costs and sizes, not measurements of RAM consumption on an identical workload.

What is known about speed. On 2,400 EverMemBench questions local ranking takes on average 11.59 ms for RAW and 13.70 ms for EPISODES; an extra 2.11 ms, about 18%. The corresponding p95 values

are 12.70 and 15.04 ms; network and generation are not included. The benchmark tables used contain no comparable measurement of end-to-end query latency for other memory systems, so speed is not compared with other systems; the duration of a full run also depends on API latency, the number of workers, retries and rate limits.

8.2 Engineering implications

Memory construction is paid before any questions; retrieval does not call the linker again. The hot view bounds the size of an indexed episode, while the cold history keeps all evidence pointers. The retrieval description can therefore be corrected without deleting the primary history. Sealing inputs and caching make it possible to separate a change of method from repeated sampling; the specification is in the appendix.

9 Risks and possible errors

Missed extraction. If no event is created, the fact remains available only through RAW. Structured output constrains the format but does not guarantee complete extraction.

Attribution and semantic overreach. A literally present quote may support only part of a description; a paraphrase may mistake the author of a message for the owner of an opinion. Therefore 100% quote validity means integrity of the address, not 100% correct facts.

False split and false merge. Free-form subject/facet wording can split one line into several episodes; retrieval can survive such a split if both records reach the context independently. An erroneous merge can bring irrelevant context into later requests. Conservative resolver decisions can reduce erroneous merges at the cost of broken chains; the frequencies of splits and merges were not measured in this work.

Context displacement. A derived hit takes a slot in the top- k and can displace a useful raw turn. This is a possible explanation of neutral or negative transfer, in particular on the speaker-conditioned GroupMemBench, but not an established cause: without gold source annotation on Group it cannot be tested directly.

10 Limitations

Two main gaps. First, the work contains no comparison with a plain paraphrase of the same messages equipped with the same sources and speaker/time fields. The delivery improvement is therefore established for the event index relative to raw retrieval, but not for the event schema relative to a simpler derived representation. Second, the temporal gain was found on a slice selected post hoc, and all of its controls use the same 300 questions; there is no independent check on a new set. Both require separate experiments.

What exactly is measured. Gold-source delivery does not mean context sufficiency or use of the support by the generator. Quote validity checks the address and the literal quote, not entailment of the derived claim; an independent human audit of events has not been completed. GroupMemBench has no gold source IDs in any of its 745 QA, so source delivery was not measured on it.

What is not yet separated. The generator receives both the retrieved messages and the derived text. A control with identical sources but without the retrieval description was prepared (B1), but no completed answers and verdicts were found. The contribution of source selection is therefore not yet separated from the interpretive hint to the generator.

Boundaries of transfer. The conclusions apply to this extractor, these budgets and bounded conversation spaces; a stronger extractor, the linking weights and the event cap were not ablated. Published rows of other systems, cost and storage are compared descriptively; they are not a controlled production TCO.

11 Conclusion

Event memory with addressable sources separates *how to search* from *what to ground on*: a self-contained event description makes short and scattered messages accessible to semantic retrieval, and immutable provenance returns the original message with its author and time.

The main result of this work is improved delivery of annotated sources. On EverMemBench the event index raises the precision of gold sources by 3.21 pp and recall by 0.64 pp, while returning fewer unique sources and more gold hits; the direction holds in all five projects and is supported on SocialMemBench. On Temporal Duration events keep an accuracy advantage (19.3% against 14.0% for RAW and HyDE) after controls for volume, order and query expansion; this is an exploratory signal that requires independent verification.

Evaluating delivery and answers separately shows the boundary of what is established: a mean accuracy gain from events and an added mean gain from linking into episodes are not established, and transfer across benchmarks is heterogeneous. For the studied retrieval scenario this gives a practical choice: separate EVENTS are a simple basis for the retrieval layer that improves source delivery, and EPISODES are an inspectable history of revisions, reaffirmations and augmentations. A strong RAW provides most of the overall level of the system, so new derived layers are best evaluated pairwise against it and on both endpoints at once — source delivery and answer quality.

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A Full EverMemBench breakdown

Type	n	RAW	EVENTS	EPISODES
Constraint	402	79.9	79.1	79.6
Multi-hop	249	12.9	13.3	13.3
Proactivity	427	65.8	64.6	64.2
Single-hop	213	89.7	89.7	87.8
Skill	169	33.7	32.0	30.8
Style	176	30.1	33.5	38.1
Temporal Duration	300	12.3	16.3	20.0
Title	196	46.4	45.9	43.4
Update	268	46.3	47.0	47.4

Table 8: Accuracy over all official categories, percent. EVENTS here is the full experiment A without token/order controls.

B GroupMemBench: absolute results by task

System	Multi-hop	Update	Ambiguity	Implicit	Temporal
BM25	40.11	25.23	14.15	40.82	54.94
Dense RAG	36.26	23.36	21.70	46.94	32.72
Hindsight	42.31	17.76	37.74	40.82	54.94
HippoRAG	39.56	27.10	30.19	42.86	29.63
A-Mem	35.16	22.43	26.42	46.94	23.46
MemGPT	22.53	17.76	20.75	28.57	12.42
Mem0	21.98	4.67	11.32	20.41	16.67
GraphRAG	12.09	14.02	19.81	14.29	5.56
RAW (ours, repaired)	37.91	28.04	25.47	48.98	37.04
EPISODES (ours, repaired)	34.07	23.36	24.53	48.98	36.42
n of our slice	182	107	106	49	162

Table 9: GroupMemBench, accuracy in percent. External rows: Table 2 of arXiv:2605.14498v1 [Yang et al., 2026]; Dense RAG uses text-embedding-3-large. Our rows are recomputed from the repaired results. Our per-type denominators are consistent with the published percentages; this does not confirm byte-identical question files or harness.

The five denominators sum to 606; the efficiency landscape in the main text is built on these non-abstention tasks. The aggregate is computed as $A_{606} = \sum_t n_t a_t / \sum_t n_t$, where a_t is a system’s published accuracy on type t and n_t the number of questions of that type in our artifacts (182, 107, 106, 49 and 162 for Multi-hop, Update, Ambiguity, Implicit and Temporal). In almost every cell $n_t a_t$ gives an integer number of correct answers, which makes our denominators consistent with the published percentages; for MemGPT on Temporal (12.42%) there is no integer, and the count is rounded to the nearest one. Thus, for Hindsight $A_{606} = 245/606 = 40.4\%$, for our RAW $210/606 = 34.7\%$, and for EPISODES $196/606 = 32.3\%$. Matching denominators do not prove a byte-identical question set, so this is a comparison of non-abstention aggregates, not a joint run. The 40.4% of Hindsight is not comparable with the 41.5% of our EPISODES on all 745 QA: these are different question sets. In the published table with abstention, Hindsight’s mean accuracy is 46.01% and BM25’s 43.22%; ours on all 745 QA is 43.09% for RAW and 41.48% for EPISODES. But abstention contains 139 questions in our artifacts against 109 consistent with the published percentages, so these four overall figures are reported separately, without a joint ranking.

The table shows why a single overall accuracy is not enough: Hindsight is stronger on Multi-hop and Ambiguity, our RAW is high on Update and Implicit, and BM25 remains a strong reference for Temporal. The difference between GroupMemBench Temporal and EverMemBench Temporal Duration also matters: the former tests order and relative time expressions, the latter the duration between event boundaries. A gap in percentages across benchmarks is not by itself an indication of model degradation.

C Evidence delivery by category

Type	n	Δ precision	95% CI
Single-hop	213	+8.28	[+6.50; +10.12]
Temporal Duration	300	+7.51	[+5.92; +9.11]
Style	176	+6.31	[+4.82; +7.84]
Update	268	+3.24	[+2.12; +4.39]
Skill	169	+2.32	[+1.04; +3.73]
Multi-hop	249	+1.63	[+0.59; +2.67]
Title	196	+1.55	[+0.85; +2.30]
Proactivity	427	+1.13	[+0.77; +1.52]
Constraint	402	+0.30	[−0.07; +0.69]

Table 10: EVENTS–RAW, paired difference in evidence precision over the nine official EverMemBench categories, percent and pp. The sign is positive everywhere; the CI excludes zero in eight categories out of nine. Compare with Table 8: the category with the largest delivery gain (Single-hop) shows no accuracy gain; its base level is already 89.7%.

D Offline editorial verification

Key micro/macro and evidence aggregates were independently checked against the per-question rows. For all 2,400 Ever questions, gold source IDs were reconstructed from the official reference spans; the intersections with expanded sources matched the stored precision/recall exactly. The table below is an additional descriptive check of the robustness of the direction, performed after the experiments, not a new registered endpoint.

Project	n	Δ precision	Δ recall
01	488	+2.50	+0.50
02	471	+3.45	+0.73
03	467	+3.22	+0.51
04	481	+3.61	+0.75
05	493	+3.28	+0.70

Table 11: EVENTS–RAW per EverMemBench project, pp.

Leaving out any single project, the mean precision gain ranges from 3.11 to 3.39 pp. This is sensitivity on this set, not confirmation of transfer to new projects. Social intervals in the main text were additionally recomputed with a network bootstrap: 10,000 resamples of the 43 networks with replacement, per-question deltas weighted by actual network size, seed 20260917.

Distribution of the number of Ever gold anchors: minimum 4, median 27, interquartile range 21–37, mean 30.12, maximum 96. A correct verdict with zero gold intersection occurs in 316 RAW, 312 EVENTS and 305 EPISODES answers. These cases mark the limits of a reference-based metric but are not in themselves an audit of the sufficiency or correctness of the answer.

When one EPISODES condition was regenerated and rescored on 300 questions, 12 verdicts changed (six in each direction), whereas the evidence metrics agreed on all questions. The equality refers to the fixed retrieval of this control, not to an arbitrary future configuration.

An additional RAW count-matched control returns as many messages as EPISODES expand on the same question: accuracy 13.3%, recall 16.6%, precision 25.1%. It also does not reproduce the observed temporal level of events. In RAW chrono relative, the original absolute timestamps were replaced by relative ones with the same set of sources: accuracy 8.3%, evidence metrics unchanged. This is a separate control of temporal formatting, not a control for removing the derived text.

The check is published in the paper sources as `verify_editorial.py`; its report `editorial_verification.json` contains hashes of the inputs read. Experiment artifacts are not modified. The extreme Wilcoxon p values of an earlier draft were removed: the old helper did not average ranks for ties; effect sizes and bootstrap intervals do not depend on that helper.

E Accuracy of token-budget matching

The stored context_tokens were compared on the same 300 QA keys, without reconstructing prompts or new generations. The counter uses o200k_base; the expanded context with block formatting is counted, but not the question or the shared instruction frame. For a pair of conditions a, b the deviation is $|T_a(q) - T_b(q)|$, and the relative deviation $100|T_a(q) - T_b(q)|/T_b(q)$. Percentiles are computed separately for absolute and relative deviations: the percentage in the table is not a ratio of two separately averaged quantities.

Pair of chrono contexts	Median	95th pct.	Maximum	Median, %
RAW — EPISODES	25	56.1	83	0.99
EVENTS — EPISODES	26	73.1	137	1.08
HyDE — EVENTS	25	55.1	97	1.06
EVENTS — RAW	35	97.0	139	1.48

Table 12: Absolute per-question token-matching deviations, in tokens except the last column. The denominator of the relative deviation is the second condition of the pair; all contexts are in chronological order.

For the key comparison EVENTS — RAW the interquartile range of the absolute difference is 16–58 tokens; the 95th percentile of the relative difference is 4.60%, the maximum 11.46%. For EVENTS — HyDE the corresponding median, 95th percentile and maximum are 25, 55.1 and 97 tokens; the relative values are 1.06%, 3.22% and 8.76%. The mean contexts of RAW, EVENTS, HyDE and EPISODES are 2458.1, 2456.1, 2451.4 and 2459.7 tokens. Matching brings budgets together but does not remove size differences on every question. These deviations are not a separate ablation of residual volume. The computation is included in `verify_editorial.py` and `editorial_verification.json`, section `token_matching`.

F Evidence–accuracy link: analysis protocol

The protocol `research/FRONTIER_MECHANISM_PREDICTION_PROTOCOL.md` fixes, before this analysis was computed, three strata outside Temporal Duration: A — EVENTS no worse on recall and precision and strictly better on at least one coordinate; B — RAW dominates in the same sense; C — a mixed result or equality. The confirmation rule requires a positive significant accuracy delta in A, a negative direction in B and an interval containing zero in C. The main condition in A is not met. The small number of discordant pairs (77, 23, 67) limits power; the CIs also admit a useful positive effect. The analysis does not establish a causal bottleneck.

Stratum	n	Δacc	95% CI	Wins/losses	McNemar
A (events dominate)	650	+0.77	[−1.85; +3.54]	41/36	.649
B (RAW dominates)	209	−2.39	[−6.70; +2.39]	9/14	.405
C (mixed)	1,241	−0.24	[−1.53; +1.05]	32/35	.807

Table 13: Offline check of the delivery–answer link on the 2,100 questions outside the discovery slice, EVENTS–RAW, percent and pp.

G GroupMemBench: restoring empty answers

The original run had 102 empty answers out of 1,490: 44 RAW and 58 EPISODES. The amendment repeats only these calls with the same sealed prompts and a larger output budget; all 102 were restored. The remaining 1,388 answers were kept byte for byte, and source IDs are unchanged. The original results were not rewritten.

The repair cost \$2.69; about 94% of the output tokens went to hidden reasoning. One shared payload was called twice while filling the cache in parallel; the final answers coincided. The repair is a post-protocol amendment, not an official leaderboard submission.

Slice	Version	n	RAW	EPISODES	Δ [95% CI]
Primary	original	150	34.0	30.0	-4.0 [-9.3; 1.3]
Primary	repair	150	37.3	33.3	-4.0 [-9.3; 1.3]
All QA	original	745	42.0	40.3	-1.7 [-4.3; 0.7]
All QA	repair	745	43.1	41.5	-1.6 [-4.0; 0.8]

Table 14: GroupMemBench before and after the repair, percent and pp. Primary: Finance+Technology, knowledge update+temporal.

All 745 official QA were checked: each contains a question identifier, its text, a reference answer and the identifier of the asking participant. This version of the questions contains neither source IDs nor evidence spans. Evidence recall/precision cannot be claimed by reconstructing gold from the answer itself without separate annotation. The hypothesis that useful sources are displaced on Group remains untested. The expanded IDs of both conditions are comparable and were found in the corpus of 120,000 messages; what is missing is precisely the gold set.

H Reproduction: models, prompts and memory scale

	EverMemBench	SocialMemBench	GroupMemBench
Networks / projects	5	43	4
Extraction sessions	3,570	348	3,153
Messages	51,023	7,355	120,000
Events after validator	16,859	639	21,407
Rejected by validator	4,548	428	6,822
Episodes	9,298	449	10,801
Attach decisions: NEW	9,298	449	10,801
AUGMENT	5,175	151	5,586
REAFFIRM	1,482	11	2,331
REVISE	904	28	2,689

Table 15: Scale of the built memory. Every stored event receives exactly one attach decision, so the decisions sum to the number of events, and the number of NEW decisions equals the number of episodes. Counts come from the sealed episodes_meta.json and attach_decisions.jsonl.

Memory models and settings. Event extraction and attach decisions are performed by gpt-4o-mini-2024-07-18 at temperature = 0, max_tokens = 8192 and JSON output. A session is passed whole, one line per message: [[turn_id]] [ANCHOR] time speaker: text. At most two events per message. Attach candidates are the three best episodes by the score $S(e, E)$ on local paraphrase-multilingual-MiniLM-L12-v2 embeddings; below a confidence of 0.5 a new episode is created; the hot view is the last five events. Answer retrieval: text-embedding-3-large (EverMemBench, GroupMemBench) and text-embedding-3-small (SocialMemBench), cosine top-10.

Document templates. Event index text: Viewpoint owner: <o>. Subject: <s>. <z>; for an episode, z is replaced by the texts of the last five events joined by ||. The generator receives the header [DERIVED EVENT / owner=<o> / subject=<s>], the event text and one [SOURCE <id>] line per source in P ; messages in L are not part of this document.

Where the prompts are and how to recompute. The extraction prompt and the attach prompt are the functions _event_extraction_prompt and _attach_prompt in reproduction/research/temporal_episode_prototype.py; the benchmark adapters are next to it. Every number in the paper is recomputed offline, without API keys:

```
python3 analysis/verify_results.py
python3 analysis/verify_evidence_endpoint.py
```

Reruns require OpenAI and OpenRouter keys; per-benchmark commands are listed in REPRODUCE.md, for example `python3 -m research.evermembench_parallel_resume --all`.

I Exact storage specification

Unit	Fields and purpose
RawTurn	ID, conversation network, speaker, timestamp, text; immutable address.
MemoryEvent	ID, index text, type, viewpoint owner, subject, temporal mode, observed time, evidence, local context.
EventEvidence	Event ID, raw source ID, literal quote; verifiable link.
EpisodeEvent	Episode ID, Event ID, order, attach operation; append-only history.
SearchDocument	kind, index text, render text, source IDs; common retrieval interface.

Table 16: The logical schema does not depend on a JSONL, SQLite or PostgreSQL backend.

J Artifacts and the boundaries of the claims

Code, protocols and configurations: <https://github.com/yaroslavlikh/temporal-episodes> (the repository is closed until publication). What it redistributes follows the licence of each benchmark: per-question predictions and verdicts for EverMemBench (data under Apache 2.0) and SocialMemBench (CC BY 4.0), and for GroupMemBench, which publishes no licence, only aggregates and SHA-256 checksums of the withheld per-question files. Model caches, embeddings and the corpora themselves are not redistributed. The identifiers and hashes of the sealed runs are listed below.

The main immutable directories:

- evermembench_temporal_episodes_official_v1_20260915_021550_MSK;
- final_sprint_A_events_full_v1_20260915_233142_MSK;
- final_sprint_B2_episodes_chrono_v1_20260915_232910_MSK;
- final_sprint_C_hyde_raw_v1_20260916_024144_MSK;
- final_sprint_C2_hyde_token_matched_v1_20260916_024306_MSK;
- socialmembench_temporal_episodes_official_harness_v1_20260912_122626_MSK.

An additional sealed GroupMemBench amendment is stored in `final_sprint_D_groupmembench_repair_v2` (1,490 rows), SHA-256:

```
results_v2      41ae844eabb292f9d0c7edbeed0476981155340ac85c5ef197d2184fcbd0d4a5
predictions_v2  20f2c60180a58d37c63f96904896266c8ac5805f83d822d321a34926850c8a96
```

The directory was not copied into a separate frozen container; its eight source files were verified against the manifest and are unchanged. Revisions used: EverMemBench code `e10b3d52f0e4`, data `a6b210a32248`; GroupMemBench `e2682e01ff49`. The figure script hashes the input summaries/reports. Publishing the code does not by itself grant permission to redistribute the benchmark corpora.

Which Social run is canonical. Two SocialMemBench pipelines are present in the artifacts, and they must not be mixed. The canonical one is `socialmembench_temporal_episodes_official_harness_v1`: `text-embedding-3-small`, a single cosine top-10 and a shared answer/judge/packing path; every number in this work comes from it, and its RAW/FLAT/VERSIONED rows are the unmodified results of the frozen run of 2026-09-11. The second, `socialmembench_temporal_episodes_full_v1`, is an earlier hybrid (local MiniLM, raw top-12 + episode top-12, a 400-word budget); its RAW gives an evidence recall of 48.5% against 67.5% for the canonical one. Comparing a condition from one run with RAW from the other produces plausible and entirely false numbers; we note this here because both directories are distributed together.

Inputs and reproduction. The stored evidence effect sizes and question-bootstrap intervals are available in `research/figures/evidence_endpoint_results.json`. The script `build_dissociation_figure.py` builds the categorical figure from these values; `build_figures.py` records hashes of the other inputs. The editorial verification uses a separate report and does not rewrite results.